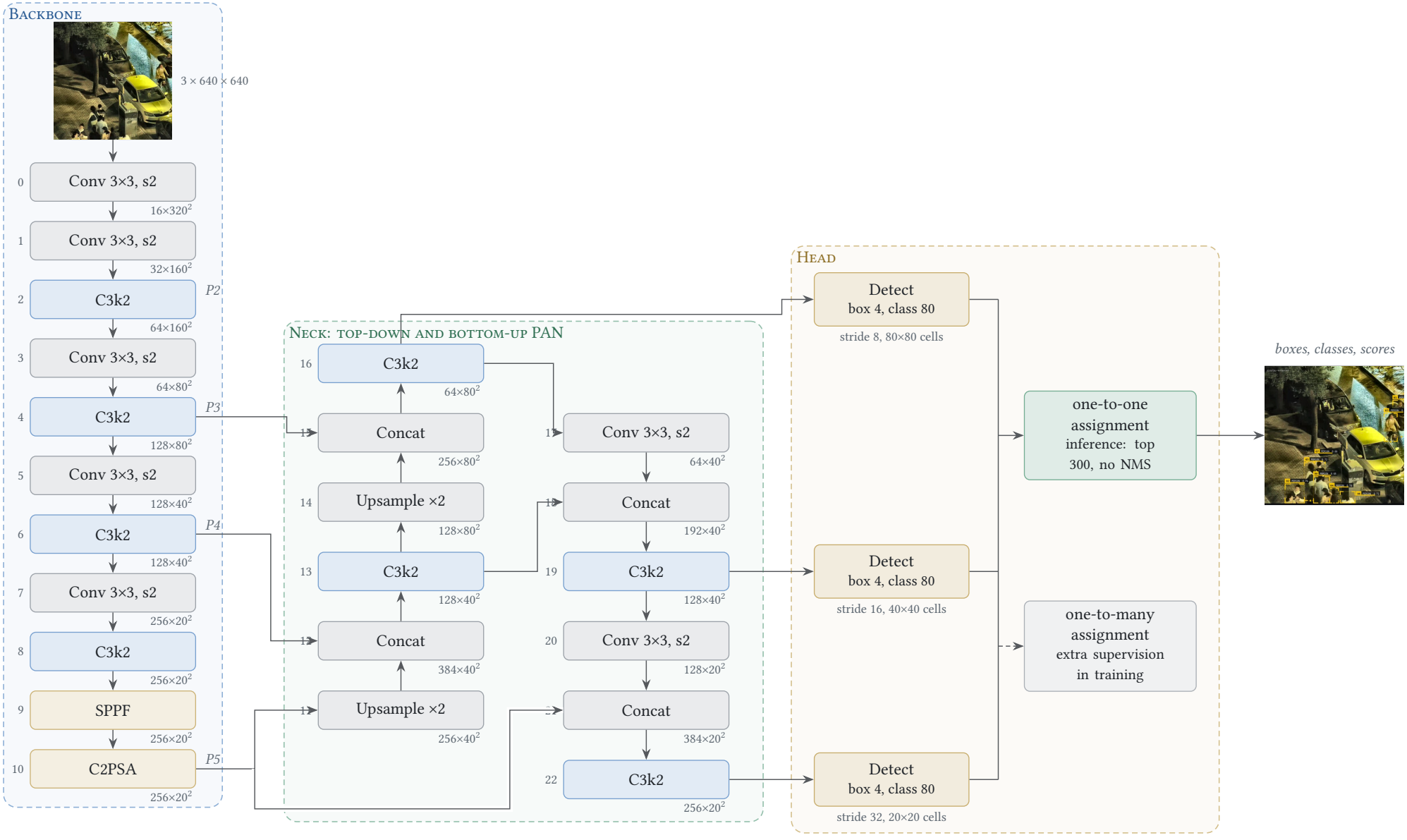

Backbone
3 × 640 × 640
0 Conv 3×3, s2
16×320²
1 Conv 3×3, s2
32×160²
2 C3k2
P2
64×160²
3 Conv 3×3, s2
64×80²
4 C3k2
P3
128×80²
5 Conv 3×3, s2
128×40²
6 C3k2
P4
128×40²
7 Conv 3×3, s2
256×20²
8 C3k2
256×20²
9 SPPF
256×20²
10 C2PSA
P5
256×20²
Neck: top-down and bottom-up PAN
16 C3k2
64×80²
15 Concat
256×80²
14 Upsample ×2
128×80²
13 C3k2
128×40²
12 Concat
384×40²
11 Upsample ×2
256×40²
17 Conv 3×3, s2
64×40²
18 Concat
192×40²
19 C3k2
128×40²
20 Conv 3×3, s2
128×20²
21 Concat
384×20²
22 C3k2
256×20²
Head
Detect
box 4, class 80
stride 8, 80×80 cells
Detect
box 4, class 80
stride 16, 40×40 cells
Detect
box 4, class 80
stride 32, 20×20 cells
one-to-one
assignment
inference: top
300, no NMS
one-to-many
assignment
extra supervision
in training
boxes, classes, scores